

Management of Metastatic Renal Clear Cell Cancer: ASCO Guideline Rapid Recommendation Update

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ASCO Rapid Recommendation Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the [ASCO Guideline Methodology Manual](#). The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options. See the Appendix for disclaimers and other important information ([Appendix 1](#) and [Appendix 2](#), online only).

ACCOMPANYING CONTENT

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Appendix

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BACKGROUND

In 2022, ASCO published a guideline on the management of metastatic renal clear cell cancer (ccRCC).¹ At that time, no recommendations for or against triplet therapy in this setting were made as that option was not available. Since then, results of the phase III COSMIC 313 trial by Choueiri et al,² which compared ipilimumab, nivolumab, and cabozantinib (IpiNivoCabo) against matched placebo with ipilimumab and nivolumab, have been published, providing the signal to update the 2022 guideline.

METHODS

The COSMIC 313 trial² is the only trial that informs this rapid recommendation update (literature search updated August 18, 2023). The original guideline Expert Panel was reconvened to review this new evidence and to review and approve the revised recommendation.

EVIDENCE REVIEW

COSMIC 313 was a phase III, double-blind trial, that randomly assigned treatment-naïve patients with intermediate or poor prognostic risk (according to the International Metastatic Renal Cell Carcinoma Database Consortium) metastatic ccRCC to cabozantinib (40 mg once daily), nivolumab (3 mg/kg of body weight, once every 3 weeks for four cycles), and ipilimumab (1 mg/kg, once every 3 weeks for four cycles) or matched placebo in addition to nivolumab and ipilimumab. Nivolumab maintenance 480 mg once every 4 weeks with either cabozantinib (experimental arm) or placebo (control arm) was continued up to 2 years. The primary end point was progression-free survival (PFS), as determined by blinded independent review according to RECIST, and was assessed in the first 550 patients who had undergone random assignment. To achieve adequate statistical power, the secondary end point of overall survival (OS) was assessed in all 855 patients who had undergone random assignment.

In the patients treated with IpiNivoCabo (median PFS not reached), the probability of PFS at 12 months was significantly higher than treatment with only nivolumab and ipilimumab (median PFS, 11.3 months; 0.57 v 0.49, respectively; hazard ratio for disease progression or death, 0.73 [95% CI, 0.57 to 0.94]; $P = .01$). Objective response rate (ORR) was also higher in the experimental group than in the control group (43% v 36%, respectively) while the complete response rate was only 3% in both arms. However, grade 3 or 4 adverse events (AEs) were more

TABLE 1. Ongoing Studies

Study Attribute	Detail
Trial ID	PDIGREE
ClinicalTrials.gov identifier	NCT03793166
Title	PD-Inhibitor (Nivolumab) and Ipilimumab Followed by Nivolumab vs. VEGF TKI Cabozantinib With Nivolumab: A Phase III Trial in Metastatic Untreated REnal Cell CancEr [PDIGREE]
Study type	Randomized, open, phase III
Disease and/or patient characteristics	Includes clear cell renal cell carcinoma
Prior treatment	No prior treatment allowed
Experimental agent	Ipilimumab and nivolumab, followed by nivolumab with cabozantinib
Comparator	Ipilimumab and nivolumab followed by nivolumab alone
Planned enrollment	1,046
Status	Recruiting
Sponsor	National Cancer Institute (NCI)
Trial ID	ICONIC
ClinicalTrials.gov identifier	NCT03866382
Title	A Phase II Study of Ipilimumab, Cabozantinib, and Nivolumab in Rare Genitourinary Cancers (ICONIC)
Study type	Open, phase II
Disease and/or patient characteristics	Renal cell carcinoma
Prior treatment	Allowed
Experimental agent	Cabozantinib, nivolumab, ipilimumab
Comparator	Single-arm trial
Planned enrollment	224
Status	Recruiting
Sponsor	National Cancer Institute (NCI)
Trial ID	NCI-2014-02379
ClinicalTrials.gov identifier	NCT02496208
Title	A Phase I Study of Cabozantinib Plus Nivolumab (CaboNivo) Alone or in Combination With Ipilimumab (CaboNivoIpi) in Patients With Advanced/Metastatic Urothelial Carcinoma and Other Genitourinary Tumors
Study type	Open, phase I dose-escalation study
Disease and/or patient characteristics	Advanced clear cell renal cell carcinoma
Prior treatment	—
Experimental agent	Cabozantinib and nivolumab with or without ipilimumab
Comparator	—
Planned enrollment	152
Status	Active, not recruiting
Sponsor	National Cancer Institute (NCI)

common in the triplet therapy group v control group (79% v 56%, respectively). Dose reductions and the need for high-dose steroids were also more common in the triplet therapy group. Follow-up for OS is ongoing. This trial was determined to have a high certainty of evidence according to the GRADE methodology.³

UPDATED RECOMMENDATION

Recommendation 3.6

Treatment with IpiNivoCabo is not recommended for patients with metastatic ccRCC. Patients interested in triplet

therapy should enroll in a clinical trial (Type: Evidence based; harms outweigh the benefits [harms of IpiNivoCabo outweigh the benefits]; Evidence quality: High; Strength of recommendation: Strong).

For all guideline recommendations, a complete summary table is available at <http://www.asco.org/genitourinary-cancer-guidelines>.

DISCUSSION

Treatment escalation remains an area of interest in renal cell cancer management, with doublet therapy largely

supplanting monotherapy.¹ The challenge of creating regimens with synergistic, or at least additive efficacy, without excessive toxicity remains paramount. Furthermore, as the components of this triplet combination are already approved, an improvement in OS along with more manageable AEs would need to be seen to demonstrate that the triplet is necessary rather than doublet or sequential treatments. COSMIC 313 noted improved PFS and ORR with its triplet regimen, but grade 3–4 AEs, including AEs of special interest, were also more common in the experimental arm. A particular challenge was related to hepatotoxicity (elevated liver function tests: AST and ALT), which can be seen with both the ipilimumab–nivolumab combination and cabozantinib

monotherapy, making attribution and management more difficult. The increased need for high-dose steroids, dose reductions, and treatment discontinuations in the triplet arm may partially explain the low complete response rate secondary to inadequate treatment exposure.⁴ Similarly, the number of patients without cytoreductive nephrectomy (kidney in place) in both arms was higher than in other pivotal trials, and readers are referred to Recommendation 2.1 on the role of cytoreductive nephrectomy from the original guideline.¹ Patients and providers who are interested in triplet therapy are encouraged to consider one of the ongoing clinical trials as the OS data from COSMIC 313 matures.^{4,5} See [Table 1](#) for a listing of ongoing studies.

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EDITOR'S NOTE

This ASCO Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at www.asco.org/genitourinary-cancer-guidelines.

EQUAL CONTRIBUTION

E.A.S. and P.J.V. were Expert Panel coauthors.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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No other potential conflicts of interest were reported.

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